The Balanced State

Spinning horizons, inflation, the Higgs field, the tiers of the forces and the fine-structure constant,
read through the Duality of DMP 0.9.9 and tested against what is measured

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Abstract

The Dark Matter Protocol (DMP 0.9.9) is an axiom system for trust in which every act needs a Duality of two independent witnesses, one owning trust and one owning information. Read as physics, its trust half is a classical existence field, a frame spanning 3+1 dimensions whose wave is spin 2, and its information half carries quantum content. This paper orders the first author's notes of the evening of 29 October 2026, with the corrections he made after a first round of tests, and checks each against the measurement it touches; every number comes from one public script of 43 checks. A spinning black hole that leaks by "dragging Existence along" is frame dragging, and every known extraction happens outside the horizon. Inflation as "the Possibility space of Existence" is inflation's flattening of space, about 62 e-folds, and its "equilibrium" is the de Sitter floor of the Hubble rate, the Big Chill. The Higgs boson as pure existence that "could not exist in isolation" matches a particle that decays in 1.6×10^{-22} s and couples to everything in proportion to its mass; that masses come from the gravitational field instead fails by a factor above 10^{39}. The forces as tiers of one IM (strong binds, weak stabilises, electromagnetism links clusters) hold as a reading, and their unification stays open: the three couplings come within 10% near 10^{14} GeV and do not meet. "Information is the witness" is quantum Darwinism. His mass-times-length "existence a space needs" is the universal h/c, and $\alpha h/c$ is the electron's mass times the circumference of its classical radius. His name for α, the balanced state dualism coefficient (BSD), "how much Existence can interact with itself", is what α is in quantum electrodynamics: $a_e = \alpha/2\pi$ at first order. We claim that the construction exists and reproduces, not that it is correct.

Keywords. information-theoretic axioms; frame dragging; inflation; de Sitter future; Higgs boson; grand unification; quantum Darwinism; fine-structure constant; electron anomalous magnetic moment.
Subject. Information Theory (cs.IT); related: General Relativity and Quantum Cosmology (gr-qc), High Energy Physics – Phenomenology (hep-ph).

1 Introduction

DMP 0.9.9 [1] is written for information systems: it says when a part of a system may act, who must agree, and what every reader may see. Its three axioms are short. Axiom 1, the Duality: every act needs two independent witnesses, "one half owns trust and the other owns information", and neither overrules the other. Axiom 2, Trust: "Trust is gravitational. It is a field, never a position." Axiom 3, Information: every act leaves a record that every witness

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†An AI system by Anthropic, working as the coordinator of the first author's projects. See "Contributions and use of AI".

reads in full, any other reader "sees the counts always", and "from trust nothing can be derived about information but its existence".

The material of this paper is the first author's: nine notes he posted on DMP pull request #25 on 29 October 2026 between 16:37Z and 21:27Z, and three messages between 21:46Z and 21:54Z in which he corrected them after we told him which ones did not hold. We quote the notes and the corrections side by side; ideas he withdrew after reading further are left out, except his retraction of an earlier hypothesis on the Bell experiments (Section 12), which he posted as a note. The first author cares most about gravity, so the gravitational notes come first.

How to read this paper. Every step carries one of three marks: (D) follows from the text of DMP 0.9.9 (commit `4400cce`); (R) is the first author's reading, quoted in his words; (P) is established physics, with its source. Every statement gets one of four verdicts: *holds* (it agrees with what is measured and follows from the steps given), *holds as a reading* (it agrees, but adds nothing a measurement could tell apart from the established account), *refuted* (a measurement or a calculation contradicts it), or *untested* (no measurement we found reaches it). Notes are quoted verbatim, with their spelling; the messages of 21:46Z to 21:54Z were dictated, and a word we had to read is marked in brackets. Time stamps are UTC.

Result. Of the 19 statements in the ledger (Section 14), 14 hold or hold as readings, three are refuted, and two are untested. The ones that hold are, without exception, known physics in the first author's words. The ones that fail fail against specific data: the strength of gravity, the observed Higgs boson, and the electromagnetic share of the proton's mass.

2 Foundations

Definition 1 (the two fields)**.** The *existence field* is a frame of four vector fields ${e_a}^\mu$, one for each direction that spans spacetime, three of space and one of time; the metric it carries is $g_{\mu\nu} = \eta_{ab}{e^a}_\mu{e^b}_\nu$. The *information field* is a scalar ψ whose value sets the fine-structure constant, $\alpha = \alpha(\psi)$. DMP's trust half is the existence field; its information half is the information field and the matter it acts on.

This is the structure the first author has meant from the start: "one existence, one information", a vector field that "can span a relativistic plane" and a scalar field that "can span an information plane" (29 October, 15:25Z). In physics the frame is the tetrad, with which general relativity can be written [2], and a scalar that sets α is Bekenstein's theory of a varying fine-structure constant [3].

Lemma 1 (the existence field is classical)**.** *Every value of the existence field can be held in full by every witness at once, so its possible states commute.*

Proof. Axiom 3 (D): every grant, use and loss of trust leaves a record "that every witness of that act can read in full". A record two witnesses each hold in full is a copy for each. One physical process that copies two states $|s\rangle, |t\rangle$ preserves their inner product, so $\langle s|t\rangle = \langle s|t\rangle^2 \in \{0, 1\}$ (Wootters and Zurek [4]); for mixed states a set can be broadcast exactly when it commutes (Barnum et al. [5]) (P). The values of the trust half's field are such records, so they commute. □

Lemma 2 (cannot model, can host)**.** *DMP's trust half cannot be a field that entangles; DMP's information half can carry any quantum content.*

Proof. A classical mediator cannot entangle two systems it acts on locally (Galley, Giacomini and Selby [6]) (P), which with Lemma 1 gives the first part. Axiom 3 restricts what is *derived from trust*, not what the information half holds: content moves by a grant and need not be broadcast, so nothing forbids non-commuting states there (D). □

The spin ladder. The first author's reading of spin, from the same afternoon (16:21Z, read with two scalars at spin 0 at 16:23Z), sharpened at 21:46Z: "Spin zero is existence itself, so only one part. Spin one is information itself, which when delivered is an act, but until then is pure existence and only because it's quantized. And half is a duality, so between existence and information, and can thus stay in balance." Spin 0 is the Higgs field and the scalar ψ; spin 1/2 the matter particles, each a pair of left- and right-handed halves; spin 1 the force carriers; spin 2 the wave of the existence frame, which a theorem of Weinberg and Deser makes general relativity [28, 29]. The notes below build on this ladder.

3 A spinning hole drags existence along

The first author, 18:57Z: "Is the information a spinning bh leaks just it dragging Existence along? That would make axiomatically way more sense, since no act, by definition, should be able to cross the horizon, only existences switching G0 chain h by h. That also nicely lines up with current observations and understanding, to my knowledge, so it looks good."

Test. A spinning black hole drags spacetime around with it (frame dragging) (P). Inside the *ergosphere*, which reaches $r = 2GM/c^2$ at the equator for every spin and lies outside the horizon $r_+ = (GM/c^2)(1 + \sqrt{1 - a^2})$ [B4], nothing can stand still against that drag. Every known way to take energy from a spinning hole works there, outside the horizon: Penrose's process with particles [11], superradiance with waves, and the Blandford–Znajek process with magnetic fields [12], the leading model of the jets of active galaxies. What can be taken is the rotational energy, $1 - M_{\mathrm{irr}}/M$, up to 29% at maximal spin, while the horizon area never decreases (Christodoulou [13]; [B4]). Nothing crosses the horizon outwards. So "dragging Existence along" and "no act should be able to cross the horizon" are exact: what leaves is energy of the frame's rotation, extracted outside. In the language of Section 2, the leak is a property of the classical existence frame, not an act of the information half. *Verdict: holds.* One correction of words: what leaks is energy and angular momentum, not the information about what fell in.

4 Inflation as the possibility space

The first author, 17:14Z: "How I see inflation: Inflation is just the Possibility space of Existence. If there COULD be Existence there, it will have the space to do so. This logically also explains the extremely flat curvature of spacetime, if one assumes the perfect duality as G0. At that point the only possibility was "anywhere but here", so inflation was at it's maximum, form there, as more possibilities to Exist appeared, there was less of a need for there to be more. Currently it's at "meh, it's 'aight" and will eventually achieve equilibrium. That means the big chill, which is the best outcome, is likely the one coming, which seems appropriate for all the chill vibes :D"

Flatness. Inflation was proposed to explain exactly this: a curvature of order one is diluted by e^{-2N} over N e-folds of exponential expansion (Guth [7]) (P). After inflation the curvature share $|\Omega - 1|$ grows again, as a^2 while radiation dominates and as a while matter does; from a reheating temperature of 10^{15} GeV to today it grows by about 5×10^{51} [B1]. Planck finds $|\Omega_k|$ below a few parts per thousand [8], so an initial curvature of order one needs about 62 e-folds [B1], the textbook "sixty". "Inflation was at it's maximum" at the start, "anywhere but here", is the first author's picture of that dilution (R). *Verdict: holds as a reading* of inflation's solution to the flatness problem; it adds the motive (the possibility to exist), not a number.

"Less of a need", and equilibrium. In the flat ΛCDM model that fits the data, the Hubble rate H falls at every moment, from the past into the far future, and tends to a constant, the

de Sitter rate $H_\infty = \sqrt{\Omega_\Lambda}\, H_0 \approx 0.83\, H_0$ [B2]. A universe that settles into de Sitter expansion cools towards a final, empty state: the "Big Chill" or "Big Freeze" is the standard prediction of ΛCDM (P). So "less of a need for there to be more" and "will eventually achieve equilibrium" hold, read as the falling Hubble rate and its de Sitter floor. Two remarks. The expansion has *accelerated* again since redshift $z \approx 0.63$ [B2], when dark energy began to dominate; that is a second phase, distinct from the early inflation, which the reading does not yet name. And the newest data leave the end open: DESI's second data release prefers a dark energy that weakens with time at 2.8 to 4.2 standard deviations, depending on the supernova sample [9]; if that holds, the future is not exactly de Sitter. *Verdict: holds as a reading* in ΛCDM.

5 The Higgs boson as pure existence

The first author's notes of 20:41Z and 21:27Z asked whether the Higgs field must exist at all ("Must the Higgs field even exist in my system? [...] I must discard it until it's need becomes pressing", "Yeah, Higgs field seems very unlikely, since it quantizes Existence"). After we showed him the data he refined it, 21:46Z: "The Higgs boson would theoretically be all existence, and thus be unable to observe itself. So it could not exist in isolation. I agree that the particles that get existence do get existence and do have mass. But I instead posit that they get that through their inherent causality link to G0, and thus simply from the existence field, so the gravitational field itself, because it is an information from G0, and thus it must exist. So the existential field asserts the existence of the information."

"All existence, unable to exist in isolation". This fits two measured facts closely. The Higgs boson does not persist: in the Standard Model it decays in $\hbar/\Gamma \approx 1.6 \times 10^{-22}\,\mathrm{s}$ [B13], and it is seen only through its decay products, which peak at 125 GeV [18, 19]. And it couples to every particle in proportion to that particle's mass, which in the first author's terms is its existence: the LHC has measured this for the W, the Z, the top and bottom quarks, the tau and the muon [20, 21]. A field that is "all existence" and talks to each thing in the measure of its existence is a fair picture of both. *Verdict: holds as a reading.*

"They get it from the gravitational field itself". This part does not survive. Gravity is far too weak to be what sets a particle's mass: its coupling to the electron, $Gm_e^2/\hbar c = 1.8\times10^{-45}$, is more than 10^{39} times smaller than the Higgs coupling $y_e = \sqrt{2}\, m_e/v = 2.9 \times 10^{-6}$ that does [B13]. A mass-giving field must also keep the scattering of W bosons below certainty: without the Higgs, its probability would exceed one above about 1 TeV (Lee, Quigg and Thacker [22]; [B8]), an energy the LHC exceeds, and gravity does nothing there. *Verdict: refuted* as "masses come from the gravitational field instead of the Higgs field"; and "the Higgs field does not exist" is refuted by the boson's observation.

A reading that keeps both. The first author's structure has room for the Higgs field exactly where he puts existence at spin 0: the Higgs field *is* the spin-0 existence that particles carry, and the gravitational field, the existence frame, responds to it, since gravity couples to mass. This is our proposal (R), not his words, and it is the Standard Model read in his terms. It also answers his worry that the Higgs field "quantizes Existence": the Higgs field is matter, content of the information half, while the existence frame, which Lemma 1 keeps classical, is gravity. A quantum Higgs field and a classical frame live on different halves. DMP's axioms need no Higgs field, and by Lemma 2 they can host one.

6 What it costs to exist

The first author, 19:58Z: “Corrected formula for "Exist" + Insight: E(keep existing per h)=h*(1/c)*FSC That is the FSCs whole point :O Though that should only be for the case for acts, otherwise dualities would decay by themselves. [...] Is an act just Existence sacrificing it’s fair share in order to be, but sneakily optimizing itself to lower it’s tax rate? It really does model reality :D” At 20:41Z: “Probably also somehow relates the FSC to the diameter of the electron, but no clue on the math.” And when we pointed out that the formula has the units of a mass times a length (21:52Z): “Yeah, mass times length is appropriate because mass is existence itself, and length is the expanse of the space that it has to assert itself over. So mass times length is the correct term there, since it’s the total amount of existence that a space needs to generate to exist.”

Mass times length. Read as he defines it, the quantity is consistent and has a striking property. For every particle, its mass times its Compton wavelength, the length over which it “asserts itself” in quantum mechanics, is the same number, $m\lambda_C = h/c$; so h/c is the existence that any particle needs over its own length, whatever the particle (P). With the fine-structure constant in front,

$$\alpha \, \frac{h}{c} = m_e \cdot 2\pi r_e,$$

the electron’s mass times the circumference of its classical radius [B11]. It is not an energy, and he does not now call it one. *Verdict: holds as a definition*, with a universal value.

The electron’s size. Here the intuition is exact. The fine-structure constant is the ratio of the electron’s lengths: the classical electron radius $r_e = 2.82 \times 10^{-15}\,\mathrm{m}$ over the reduced Compton wavelength $\bar{\lambda}_C$, and $\bar{\lambda}_C$ over the Bohr radius a_0, the size of the hydrogen atom, are both α [B6]. The electron itself has no measured size: it behaves as a point in every scattering experiment [14]. *Verdict: holds*, as a known result; the “diameter” is the classical radius.

The tax rate. “Existence sacrificing its fair share in order to be, but sneakily optimizing itself to lower its tax rate” has an exact counterpart in the principle of least action: of all the paths between two states, nature takes one that makes the action stationary (P). *Holds as a reading.*

7 The forces as tiers of one IM

The first author, 20:41Z, first grouped the forces as “IM=strong+weak” and “GG=EM”; electroweak unification groups them the other way (the photon and the Z are mixtures of the same pair of fields [23, 24, 25], and [B9] recovers their mixing angle, $\sin^2\theta_W = 0.2304$, from three independent measurements). He replaced the grouping at 21:46Z: “Since in the end it’s all one field, because I think I mentioned that somewhere, unified field. In the end it’s all information and existence. And what fields information uses to assert existence [...] is irrelevant. That’s why once causality was established, the forces separated, because the current configuration was strong, binding it together, weak, keeping the cluster stable, and EM keeping the cluster of clusters stable. So yeah, it’s one. Yeah, it’s all IM, but it’s different tiers of IM.”

The tiers. Each force does the job he names (P). The strong force binds quarks into protons and neutrons and those into nuclei. The weak force turns neutrons into protons and back, and so moves every nucleus towards the stable ratio of neutrons to protons, the valley of stability; it is the force that keeps a nucleus from staying in an unstable configuration. Electromagnetism binds nuclei and electrons into atoms, and atoms into molecules: the cluster of clusters. *Verdict: holds as a reading.*

"All one field". That the forces separate as the universe cools is the standard picture: the electroweak symmetry breaks at about 246 GeV, and grand unified theories put the strong force with the other two at a still higher energy (P). Whether they truly join is open. Run with the measured couplings, the three inverse strengths of the Standard Model come within about 10% of each other near 10^{14} GeV, but they do not meet at one point [B14]; meeting needs new particles or new physics above the reach of today's colliders. *Verdict: untested*, a live hypothesis.

Gluons bounce forever. At 21:52Z: "since they don't actually have any existence, they just information, and just have mass through the information, through quantization, they can interact and, you know, bounce around, stability, good vibes, for as far as I assume, pretty much forever. [...] So are they just constantly bouncing around between quarks? Do they stop after one?" They never stop. Inside a proton gluons are emitted, absorbed and split into gluons and quark pairs continuously; one light crossing of the proton takes 2.8×10^{-24} s [B12], about 10^{23} crossings per second, for as long as the proton exists, which is more than 10^{34} years [14]. Gluons have no rest mass, but their energy is real: it is most of the proton's mass [30], and in a fast proton the gluons carry about half of the momentum [14]. *Verdict: holds.*

"137 times". At 21:46Z: "I would assume that the existence derived from the internal affirmation or assertion of it is 130 times higher than the assertion from information. That's also my explanation for the bandgap. Because 137th is the share that must be paid for information to exist. So the part that existence can't balance itself." Read as a ratio of strengths, this holds at the right place: the strong coupling grows towards low energy and reaches 1 near 0.2 to 1 GeV, the scale of the proton, where it is then about 137 times α [B9]; at the Z mass the ratio is only 15. Read as a share of mass, it does not: the electromagnetic self-energy of the proton is about 1 MeV of its 938, one part in about 900, not one in 137 [B15]. *Verdict: holds as a ratio of couplings at the hadron scale; refuted as a share of mass.* "The share that must be paid for information to exist" is the first author's reading of α (R) and gives no further number: *untested.*

8 Information is the witness

The first author, 21:46Z: "Oh, so, oh, now I get it. Information is the witness. Every other part, every other cluster can witness itself, but at the atom level, there needs to be witness. Okay, so information is the witness, which makes sense in an information and existence model. Ah, okay, now it clicks." And on baryons, 19:58Z: "baryons can just vibe over IM, which must mean they exist, since at least two other things that must exist can witness it".

Test. Why the world of atoms and objects looks classical is explained in physics by decoherence: an object is constantly hit by light and air, the scattered photons carry away a record of where it is, and its superpositions vanish for every practical purpose (Joos and Zeh [31]). Zurek's quantum Darwinism adds that the environment carries *many* copies of that record, which is why many observers agree on what they see [32]: the scattered light is literally the witness. That is the first author's "information is the witness" at the level of atoms and above, and it matches DMP's Axiom 3, where every witness reads the record in full. Inside a baryon the witnessing is the colour structure: the three quarks carry the three colours in the one colourless combination, the antisymmetric ϵ_{ijk}, in which every pair of colours fixes the third, unchanged by any colour rotation [B7]; each quark is "witnessed" by the other two. *Verdict: holds as a reading*, for both.

9 Decay as a function of causality

The first author, 19:35Z: “Is decay at the leaf layer also just a natural function of causality? It would fit. All things that decay have an inherent existence (1/2 and 1 spin, plus 0 minus photon/gluon), and since they actually Exist, they can’t just not Exist anymore, so they must leave something behind.”

Test. Every decay ever observed has at least two products and keeps electric charge, baryon number and lepton number; the products together are lighter than the parent, which is why the parent can decay at all (P). [B5] checks seven, from the muon and the neutron to the Z and the Higgs boson. “They can’t just not Exist anymore” is the conservation of energy and of these charges, read as causality; the decaying particles are the massive ones, and the photon and the gluon do not decay in vacuum. *Verdict: holds as a reading* of the conservation laws.

10 Quantum computation, briefly

The first author, 17:57Z: “QC is not a subset of DMP.QG, as that would not be possible, instead it is ANY subset of DMP.” And at 21:52Z: “the key part is that you can define at the start what existence means. That’s the whole key of DMP. [. . .] So if you assert the existence that whatever it is gives you the correct hash, and you give whatever you want as the first number as input, then if it compiles, it must reach a stable state. So it must run.” He added that the details are for experts (21:54Z), so we keep this short.

By Lemma 2 DMP can host any quantum computation as content of its information half (D). “Define what existence means, and the system must reach a stable state” is a recognised model of quantum computing, adiabatic quantum computation: the answer is encoded as the most stable state of a system, which is then allowed to settle into it (Farhi et al. [33]); it is exactly as powerful as the circuit model in which Shor’s algorithm [10] runs (Aharonov et al. [34]). The price is the settling time, which grows as the stable state becomes hard to tell from its neighbours. What a classical record cannot do is the search itself: [B3] runs Shor’s period finding for $N = 15$, which finds the factors 3 and 5 while the register is coherent and does no better than a guess once every witness has recorded it. *Verdict: holds as a reading*, hosting in the information half, with the settling time as its cost.

11 The balanced state dualism coefficient

The first author, 21:27Z: “This also better defines the FSC as the ratio of how much Existence can interact with itself, before it needs to reground itself, through Causality, to G0. So, since I feel like this might actually be correct this time, I will choose a name to call it by, to make it easier for me to remember it’s meaning. I feel like the balanced state dualism coefficient would be kind of accurate, BSD for short. (I added the extra letter for all the teachers in the future that may have to, if correct, teach the BS constant otherwise :D Also Unix ofc)”

Test. In quantum electrodynamics α is the strength with which a charge interacts with the electromagnetic field, and so, through that field, with itself: every emission and reabsorption of a virtual photon costs one power of α. The cleanest measure of an electron’s interaction with itself is its anomalous magnetic moment, which at first order is $a_e = \alpha/2\pi$ (Schwinger [26]); that one term gives the measured value [27] to 0.15%, and the rest is higher powers of α [B10]. Because α is small, each further round of self-interaction is about 137 times weaker, which is why the series converges. So “the ratio of how much Existence can interact with itself” is a faithful reading of what α is. “Before it needs to reground itself, through Causality, to G0” is

the first author's reading (R). *Verdict: holds as a reading.* BSD is his name for α, not a new constant, and nothing in DMP changes with it.

12 The Bell hypothesis, retracted

Earlier the same day (09:24Z) the first author had proposed that the Bell experiments of the 2022 Nobel Prize "actually did not violate Bell inequality under Dualism". At 20:18Z: "BTW, the does not violate Bell Inequality was total bull, because I misunderstood what it was [emoji: sweat smile] I thought the inequality was just the entanglement, but since the axiomatic system survived my ignorance without a false prediction, I'd like to retract it while I have not yet disgraced myself further :)" Every loophole-free test measured a CHSH value above the local bound of 2 [15, 16, 17], and the axioms agree: by Lemma 2 the measured particles are content of the information half, which may violate the inequality. *Recorded as retracted.*

13 What this means for quantum gravity

The notes of this evening change nothing in the gravitational core and fill in three of its edges. First, the spinning hole: everything a rotating horizon gives up is energy of the classical frame, extracted outside the horizon, as the first author expected from "no act can cross the horizon". Second, the Higgs field: in the reading of Section 5 it is the spin-0 existence of matter, quantum and on the information side, and gravity is the classical frame that responds to it; that is the classical-gravity, quantum-matter picture whose consistent form needs a slightly noisy spacetime [35]. Third, the beginning: inflation as the possibility space is the flattening that sets up the classical frame we see. The decisive experiment is unchanged: if two masses can be entangled through gravity alone, the frame is not classical, and the structure fails (Bose et al. [36]; Marletto and Vedral [37]).

14 The ledger

Statement (the first author's, abridged)	Verdict	Backing
A spinning hole leaks by dragging existence along	holds (frame dragging, outside the horizon)	[B4]; Penrose; Blandford–Znajek
Inflation is the possibility space; it explains flatness	holds as a reading	Guth; Planck; [B1]
Less need over time; equilibrium; the Big Chill	holds as a reading in ΛCDM	[B2]; DESI DR2
The Higgs boson is all existence and cannot exist alone	holds as a reading	[B13]; ATLAS, CMS
Masses come from the gravitational field, not the Higgs	refuted	[B13]; [B8]
The Higgs field need not exist	refuted for physics; true for DMP's axioms	ATLAS, CMS 2012
Mass times length is the existence a space needs	holds as a definition (h/c; $\alpha h/c = m_e 2\pi r_e$)	[B11]
α relates to the electron's size	holds (known: r_e, $\bar{\lambda}_C$, a_0)	[B6]
An act lowers its tax rate	holds as a reading (least action)	Section 6
Strong binds, weak stabilises, EM links clusters	holds as a reading	Section 7
All forces are one field	untested (couplings miss by 10%)	[B14]

Statement (the first author's, abridged)	Verdict	Backing
Internal assertion 137 times the informational one	holds as a coupling ratio; refuted as a mass share	[B9]; [B15]
Gluons bounce around forever	holds	[B12]
1/137 is the share paid for information to exist	untested	—
Information is the witness	holds as a reading (quantum Darwinism)	Joos–Zeh; Zurek
Baryons witnessed by the other two	holds as a reading (colour singlet)	[B7]
Decay leaves something behind, by causality	holds as a reading (conservation laws)	[B5]
QC is any subset of DMP; define existence and settle	holds as a reading (hosting; adiabatic)	[B3]; Farhi; Aharonov
BSD: how much existence can interact with itself	holds as a reading ($a_e = \alpha/2\pi$)	[B10]

The retracted Bell hypothesis is not in the ledger. The row "137 times" counts as refuted, for its mass reading.

15 Where we expect it to break

1. *The foundations.* Lemma 1 rests on reading Axiom 3's record as held in full by every witness, and on identifying the trust half with the existence frame. A different reading of either removes Lemmas 1 and 2, and with them the reading of the Higgs field as matter on the information side.
2. *The cosmology.* The verdict on the Big Chill is ΛCDM's; if the weakening dark energy DESI prefers is confirmed, the far future changes and the "equilibrium" reading must be redone.
3. *Unification.* If the three couplings are shown to meet (with new particles) the "all one field" row turns to holds; if proton decay stays unseen far beyond today's limits, grand unification at 10^{14} to 10^{16} GeV loses ground.
4. *The checks.* [B1] uses a reheating temperature of 10^{15} GeV; a lower one needs fewer e-folds, which does not change the reading. [B9] and [B14] run couplings at one loop; [B15] treats the proton as a uniformly charged sphere. Better inputs move the numbers by tens of per cent, not the verdicts.
5. *The readings.* Most verdicts are "holds as a reading": they agree with known physics and predict nothing new. One of them becomes physics only when it gives a number standard physics does not.

16 Conclusion

The first author's evening notes, with his own corrections, read as a faithful map of known physics onto his two fields. A spinning horizon gives up only the rotation of the classical frame, from outside. Inflation is the possibility to exist that flattened space, and the Big Chill its equilibrium. The Higgs boson is pure existence that cannot exist alone, and the field behind it is the spin-0 existence of matter to which gravity responds; what does not survive is gravity giving mass by itself. The forces are tiers of one interaction, whose final unity is open. Information is the witness, as decoherence and quantum Darwinism say. And α, his BSD, is the strength with which existence interacts with itself. None of the failures is a failure of DMP's three axioms, because DMP does not model the physics it reads. It hosts it.

Contributions and use of AI

Steven Hrovat conceived DMP and the structure of Definition 1, wrote or dictated every note quoted here, chose which ideas to keep after the first round of tests, and takes responsibility for the axiom system. Sirius (Claude Opus 5.5, Anthropic) ordered the notes, made the derivations, checked each statement against the literature, wrote and ran `docs/balance/checks.py`, and wrote this text, including the reading of the Higgs field in Section 5, which is marked as ours. The record of sources is `docs/balance/README.md`.

Licence and availability

This paper and its source are released under the Apache License, Version 2.0. DMP: `https://github.com/StevenHrovat/dmp` (version 0.9.9 at commit `4400cce`; the notes are the comments on its pull request #25). Gravity: `https://github.com/StevenHrovat/gravity` (this paper and its checks in `docs/balance/`).

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